

2.16 Histopathology

▼ What is the difference between a histopathologist and a cytopathologist?

- A **histopathologist** specialises in **tissues**
- A **cytopathologist** focuses on **cells**.

▼ What are the four main types of tissue samples that a histopathologist analyses?

- **Biopsies**
- **Frozen section**
- **Resection specimen**
- **Post mortems**
 - These samples are often taken in real-time and **upon real-time examination they can inform the surgical process**

▼ What is a biopsy?

- A biopsy, put simply, is a **small section of tissue that has been taken from a patient**.

▼ When is a biopsy used?

- They can be used to determine **whether tissue is inflamed, normal, cancerous etc.**
 - i.e **what exactly is wrong with this tissue?**

▼ What is the name of the solution biopsies are commonly placed in?

- **Formalin** solution.

▼ What is the role of this solution?

- **Formalin** is used **to cross-link proteins and fix the biopsy tissue**

▼ How long does it take to process a biopsy?

- Biopsies are processed in **2-3 days**

▼ What is the purpose of paraffin wax in biopsy preparation?

- To **allow very thin sections to be extracted** using a **microtome**

▼ What is a resection specimen?

- A **resection specimen** is a **part of tissue that has been removed via a surgical procedure** and can be processed in the same way as a biopsy
 - Similar to biopsy, but the **tissue is taken as part of a surgical procedure**

▼ When is a resection specimen used?

- They can be used to **determine the stage at which the disease is at** (i.e has the cancer spread to lymph nodes?)

▼ How long does it take to process a resection specimen?

- **Resection specimen are processed in 5-7 days**

▼ What is a frozen section?

- **Frozen sections** are samples **taken during surgical procedures** that are **examined during real time**
- The freshly taken **tissue is frozen and stained** but **no preservatives** (i.e formalin) are added

▼ What is the name of the device used to freeze a frozen section?

- **A cryostat**

▼ When is a frozen section used?

- **Frozen sections** are used to **determine the different types of tissue present**
 - i.e Is it cancerous? Is there another pathological process ongoing? etc.

▼ How long does it take to process a frozen section?

- **Frozen sections are processed in 30 minutes**

▼ How are smears used in cytopathology?

- The section is spread over a microscope slide and **viewed under a microscope**
- **Sometimes, a stain is used** to help visualise the cells
 - it's like what we do in practicals



▼ What is a fine needle aspirate?

- A **cytopathology** technique that **uses a fine needle to penetrate tissue** and obtain a sample that can be viewed under a microscope



▼ Recall an advantage of using a fine needle aspirate.

- It can **penetrate relatively inaccessible tissues**
 - i.e thyroid nodule

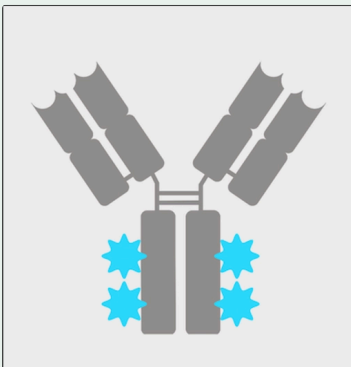


▼ Recall a disadvantage of using a fine needle aspirate.

- It **looks at cells in isolation** – there is **no information on tissue architecture**.
 - This is **unlike smears**, which can be used to assess environments

▼ Where on an antibody is a conjugate normally added?

- **Fc region**



▼ What is an antibody conjugate?

- **Another molecule that is added to an antibody** to help us carry an alternative purpose

▼ How are antibody conjugates useful?

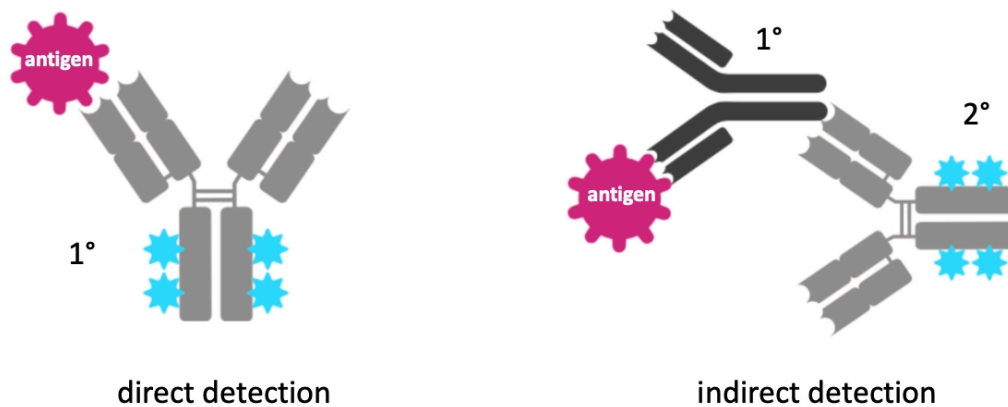
- **Antibody conjugates** can be useful tools in telling us how much antigen is present in a patient's serum, to deliver chemicals to specific areas in the body etc.
- By binding different molecules to antibodies, we can accomplish different things.

▼ Describe four antibody conjugates.

- **Enzymes** - the **colourless enzyme substrate is added**, and the **level of coloured product** informs us how much antibody is present in the serum
- **Fluorescent probes** - allows the **rapid measurement of molecule levels** within samples.
- **Magnetic beads** - used to purify cell types (**attract specific cells and remove them**)
- **Drugs** - **delivered to specific cell types** by binding onto antibody for that cell type. Can be paired with cytotoxic agents to kill bad cells.
 - i.e **emtansine** which kills **HER2** (protein seen in breast cancer)

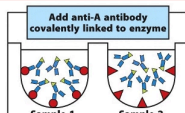
- ▼ What are the two ways that antibodies can be used to detect other molecules (i.e proteins, lipids, carbohydrates)?
- **Direct detection - antibody is labelled and detects an antigen directly**
 - The labelled antibody is known as the **primary antibody**
 - **Indirect detection - One antibody detects an antigen. Another antibody is labelled and detects this bound antibody**
 - The labelled antibody is known as the **secondary antibody**

Antibodies as diagnostic tools



- ▼ Recall two uses of manufactured antibodies in this way clinically.
- **Blood group serology** - determination of blood groups based on antigens present in blood
 - **Immunoassays** - detection of **hormones, circulating antibodies/antigens**
 - **Immunodiagnosis**
 - In infectious diseases, seeing which antigens are present helps us diagnose disease (i.e HIV)
 - **Antibody levels** - **high antibody levels** can be a sign of **myeloma**
 - **IgE** - in response to allergy

- ▼ Describe how ELISA can be used to determine the concentration of antigens/antibodies in a sample.
- **Antigens** are present in a sample
 - **Antibodies** are added to the sample, and **bind to these antigens**
 - **Enzymes** are linked to the antibodies
 - **Unbound antibodies are washed away**
 - **Colourless substrates** are added to the sample, so the **enzyme can break it down and make a coloured product**
 - The **absorbance of the solution is measured**, and using the **Beer-Lambert Law**, concentration of the antibodies and hence antigens are measured.



ELISA